

Osmar Alejandro Liera Gómez

Applied AI Engineer · LLM Agents & Production AI Integration

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PROFESSIONAL SUMMARY

Software engineer focused on **putting AI to work inside real business systems**. I design and ship **LLM agents with tool calling, structured outputs and human-in-the-loop approval**, and integrate them with enterprise data and APIs. At ELROI Labs I built the AI layer of our CRM: an **MCP server exposing 23 tools** that lets AI assistants query the pipeline, plan follow-ups and act on records under the user's own permissions, with every action audited. At Tangramx I work daily in **Python (FastAPI)**, where I shipped **LLM-based OCR to production** and **evaluated it against ground truth** before rollout. I also teach applied AI for business at the Instituto Tecnológico de La Paz.

TECHNICAL SKILLS

Agents	Tool calling, MCP servers (tools, prompts, resources), human-in-the-loop approval, conversation state	LLMs	OpenAI (Vision, Responses API), structured outputs via JSON Schema, prompt design & guardrails
Evaluation	Ground-truth validation, deterministic post-processing of model output, data-quality analysis	Python	FastAPI, SQLAlchemy, Alembic, REST APIs, JWT + 2FA
Integration	REST APIs, OpenAI-compatible model gateways, WhatsApp messaging API, Shopify, Magento	Data	SQL, MySQL, SQLite, data modeling, versioned migrations, Google Document AI
Also	Laravel (PHP), React 18, TypeScript, Tailwind CSS, automated feature testing	Cloud / Ops	Azure, Docker, Linux, Git/GitHub, CI/CD
Languages	Spanish (native) · English intermediate (B1–B2): daily technical reading and written communication		

PROFESSIONAL EXPERIENCE

Founder & Technical Lead

ELROI Labs — La Paz, BCS

Present

Software, automation and applied-AI studio. I lead technical operations and deliver projects end to end.

OS CRM — AI agent layer

Internal CRM for leads, pipeline, multichannel follow-up, clients and projects (Laravel, React)

- Built an **MCP server with 23 tools, 3 prompts and a playbook resource** so AI assistants (Claude and others) can search leads, plan the day, log follow-ups, schedule tasks, import leads and generate pipeline and activity reports.
- Agents run with **the token owner's permissions**, every change is written to the audit log, and outbound WhatsApp messages require **explicit human approval** before sending.
- Developed **LLM agents with structured outputs** (JSON Schema): a lead-contact assistant that drafts messages, call scripts and next tasks; a project planner that breaks scope into tasks with acceptance criteria and estimates; and a WhatsApp writer grounded in the real conversation and an editable business context.
- Designed **guardrails against hallucination**: answer only from supplied data, ask instead of assuming, no prices or commitments, per-agent temperature and token limits.
- Wrote a **custom OpenAI-compatible model gateway** with model-catalog discovery, caching and fallback; multi-provider setup (OpenAI, Anthropic, Gemini, Azure and others).
- Covered the MCP server, AI assistant and WhatsApp flows with **automated feature tests**.

Client work

- Applied AI** for customer service, lead prequalification and follow-up automation.
- Integrations on **Shopify and Magento** syncing catalogs, inventory, orders and customers; professional contribution to SHASA Mexico, SHASA USA and Hypsters.

Full-Stack Developer

Tangramx

Present

Management applications for insurers, healthcare institutions and logistics operators. Primary stack: Python / FastAPI.

Vitase / Paciente Integral — LLM-based document OCR

Medical records and claims management for insurers (Jalisco Health Services)

- Integrated **OpenAI Vision** for OCR of Mexican national ID cards (INE), replacing the previous engine.
- Built a **deterministic validation layer on top of the model output** for CURP: check-digit reconstruction, homoclave recovery and edge cases (Ñ, MARÍA/JOSÉ compound names).
- Evaluated against ground truth**: 25 real ID cards with **100% name accuracy**; documented the data-quality findings that guided rollout.

ATC — Account Statement Processor

Automated processing of medical account statements (FastAPI)

- Fixed incorrect discount detection and an off-by-one date shift in the parser; wrote the local setup guides.

Compass WMS · ATCSMART

Warehouse management and insurance-claims systems

- Implemented **true server-side pagination**, fixed a **race condition** in multi-file upload, and added Excel exports segmented by insurer and hospital.

TEACHING & TRAINING

Instructor — Applied AI for Business

Present

Instituto Tecnológico de La Paz

- Training on the **practical use of AI in business**: process automation, operational productivity and adoption of AI tools.
- Training on **applying AI to social media** and content generation.

PERSONAL PRODUCTS

PMCRM

Workspace for managing projects, files, notes, clients and collaboration from a single platform.

AVT

Multitenant SaaS for land sales: users, plots, zones, maps and commercial operations.

NopalGreen

Operations platform for tortilla producers: POS, production, inventory, delivery and reporting.

Based in La Paz, BCS · open to remote and hybrid opportunities.

64 public repositories · osmarlg.elroi.cloud